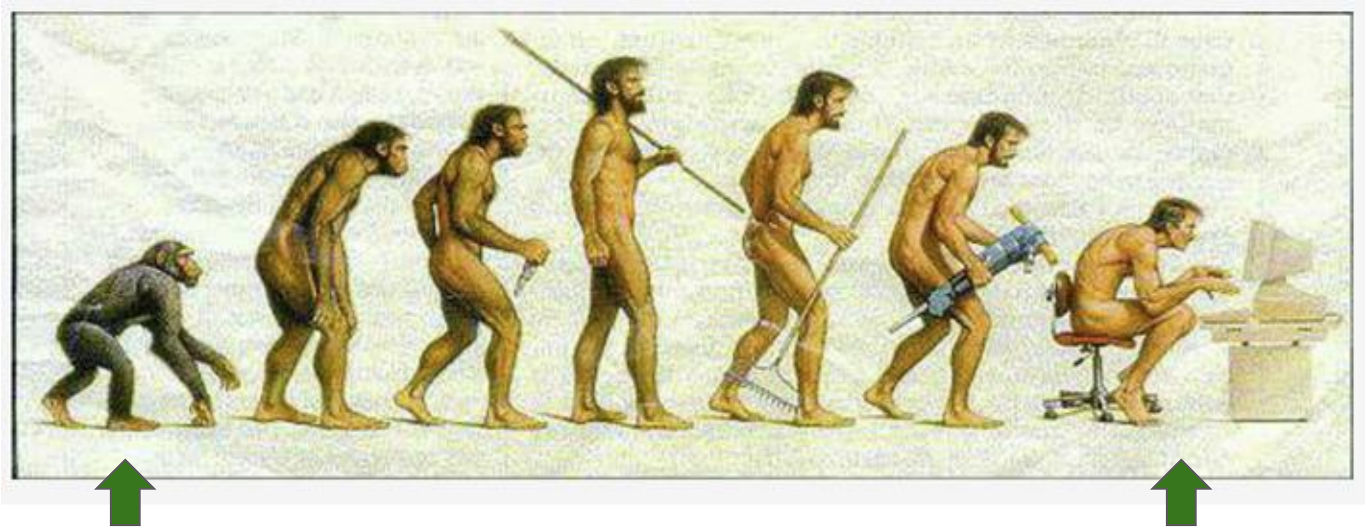
Exploration of Chimpanzee isoform diversity

2026 hackathon Group3



Where we started

Where we now

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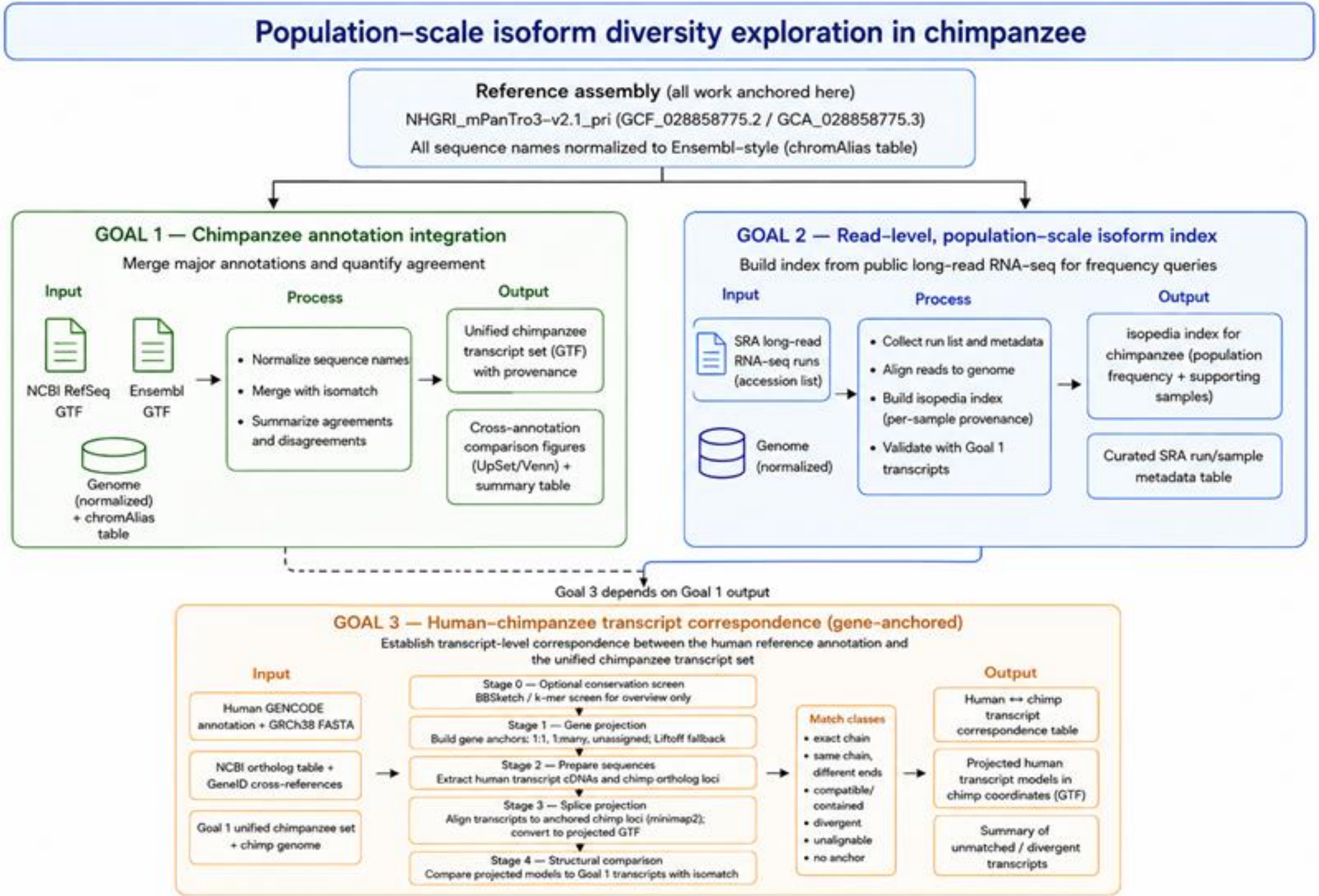
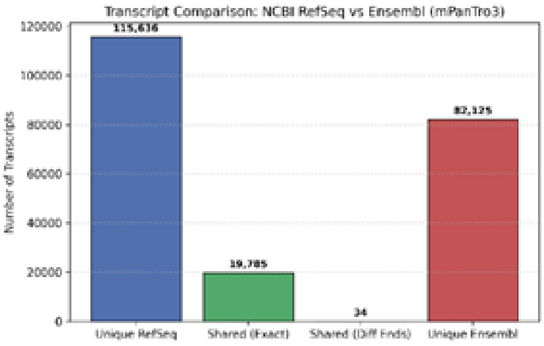


Population-scale isoform diversity exploration in chimpanzee
Reference assembly (all work anchored here)
NHGRI_mPanTro3-v2.1_pri (GCF_028858775.2 / GCA_028858775.3)
All sequence names normalized to Ensembl-style (chromAlias table)
GOAL 1 — Chimpanzee annotation integration
Merge major annotations and quantify agreement
Input
NCBI RefSeq GTF
Ensembl GTF
Genome (normalized) + chromAlias table
Process
Normalize sequence names
Merge with isomatch
Summarize agreements and disagreements
Output
Unified chimpanzee transcript set (GTF) with provenance
Cross-annotation comparison figures (UpSet/Venn) + summary table
GOAL 2 — Read-level, population-scale isoform index
Build index from public long-read RNA-seq for frequency queries
Input
SRA long-read RNA-seq runs (accession list)
Genome (normalized)
Process
Collect run list and metadata
Align reads to genome
Build isopedia index (per-sample provenance)
Validate with Goal 1 transcripts
Output
isopedia index for chimpanzee (population frequency + supporting samples)
Curated SRA run/sample metadata table
Goal 3 depends on Goal 1 output
GOAL 3 — Human-chimpanzee transcript correspondence (gene-anchored)
Establish transcript-level correspondence between the human reference annotation and the unified chimpanzee transcript set
Input
Human GENCODE annotation + GRCh38 FASTA
NCBI ortholog table + GeneID cross-references
Goal 1 unified chimpanzee set + chimp genome
Stage 0 — Optional conservation screen
BBSketch / k-mer screen for overview only
Stage 1 — Gene projection
Build gene anchors: 1:1, 1:many, unassigned; Liftoff fallback
Stage 2 — Prepare sequences
Extract human transcript cDNAs and chimp ortholog loci
Stage 3 — Splice projection
Align transcripts to anchored chimp loci (minimap2); convert to projected GTF
Stage 4 — Structural comparison
Compare projected models to Goal 1 transcripts with isomatch
Match classes
exact chain
same chain, different ends
compatible/ contained
divergent
unalignable
no anchor
Output
Human ↔ chimp transcript correspondence table
Projected human transcript models in chimp coordinates (GTF)
Summary of unmatched / divergent transcripts

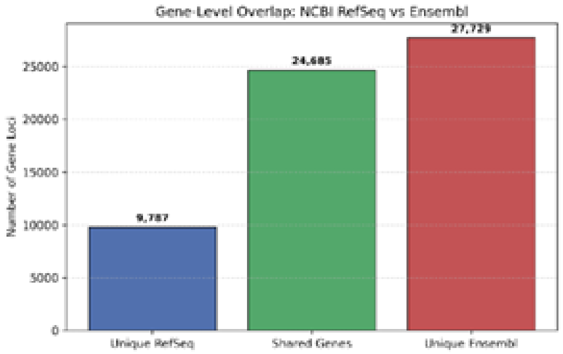
Transcript level comparison

Transcript annotation metric	Count
Total RefSeq transcripts	135,455
— Multi-exon	128,338
— Mono-exon	7,117
Total Ensembl transcripts	101,944
— Multi-exon	77,218
— Mono-exon	24,726
Shared transcripts	19,819
— Exact splice & boundary match	19,785
— Structural match / boundary difference	34
Unique to RefSeq	115,636
Unique to Ensembl	82,125



Gene level comparison

Gene annotation metric	Count
Total RefSeq genes	34,472
Total Ensembl genes	52,414
Shared genes (overlapping loci)	24,685
Unique to RefSeq	9,787
Unique to Ensembl	27,729



What causes the difference?

Biotype of unique Ensembl genes	Count	Biotype of unique RefSeq genes	Count
lncRNA	18,557	lncRNA	6,155
misc_RNA	5,384	protein_coding	2,834
protein_coding	1,694	tRNA	505
Y_RNA	751	miRNA	173
snRNA	443	transcribed_pseudogene	43
miRNA	432	V_segment	26
snoRNA	224	misc_RNA	24
IG_V_gene	60	rRNA	20
TR_V_gene	58	C_region	4
pseudogene	48	snRNA	3
rRNA	33		
TR_J_gene	24		
vault_RNA	9		
processed_pseudogene	6		
ribozyme	4		
TR_C_gene	2		

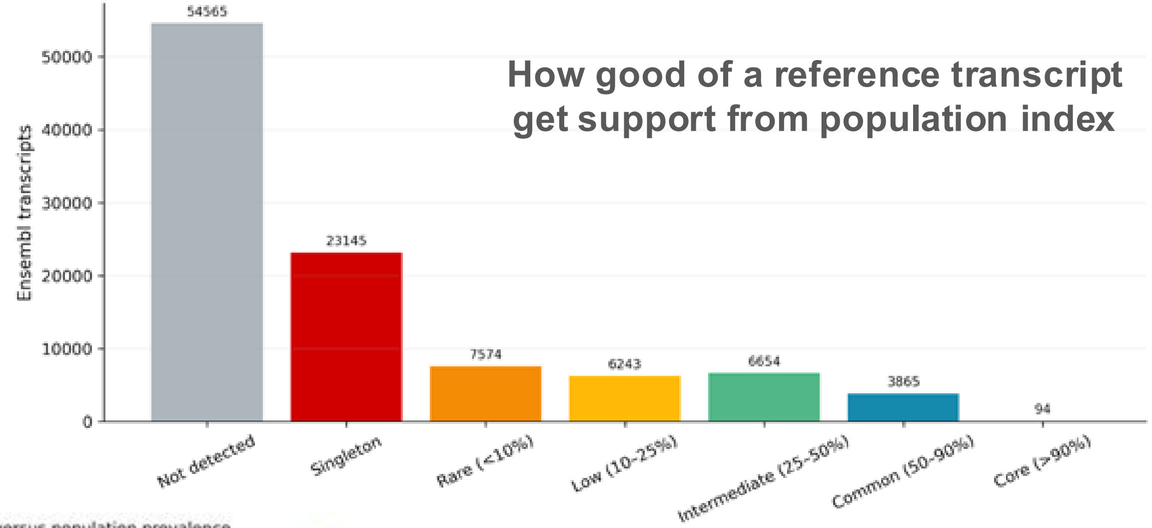
The unified transcript set

`merged_annotation.gtf.merged.gtf.gz,`
`merged_annotation.gtf.track.tsv.gz`

Transcript found evidence in each sample

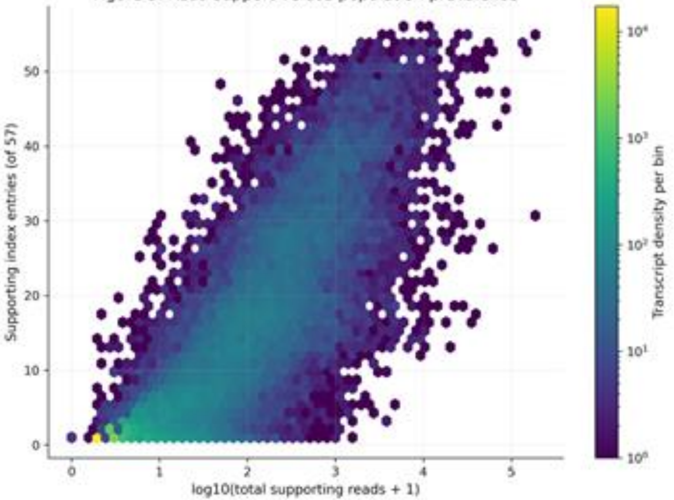
Figure 4. Isopedia expression support per sample

Figure 8. Delivered 57-entry frequency classes



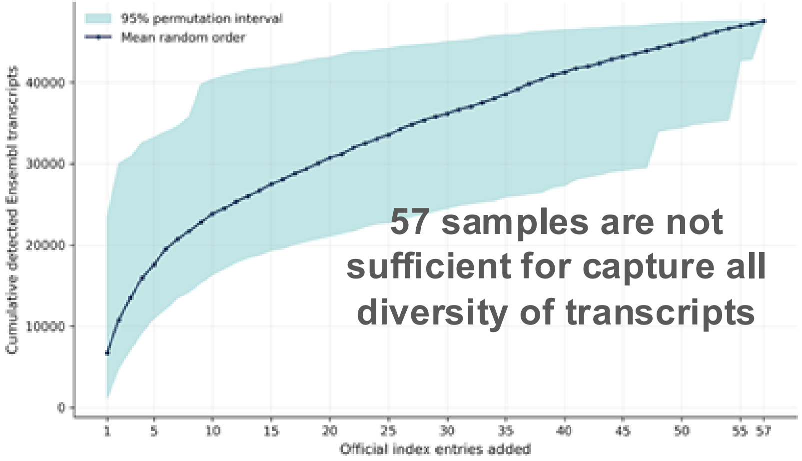
How good of a reference transcript get support from population index

Figure 9. Read support versus population prevalence



Strong correlation between population frequency and expression level

Figure 13. Delivered 57-entry discovery curve



57 samples are not sufficient for capture all diversity of transcripts

- **78,733 total human genes** in GENCODE v50.
- **20,107 protein-coding genes** in GENCODE v50.
- **22.6% of protein-coding genes** were linked to an Entrez GeneID using GENCODE metadata and NCBI Ensembl cross-references.
- NCBI chimpanzee RefSeq ortholog assignments (**GCF_028858775.2, RS_2026_05**) provided **17,903 human–chimpanzee ortholog pairs**.
- All identified ortholog pairs were **strictly one-to-one**.
- Overall, **17,869 of 20,107 protein-coding genes (88.9%)** were anchored to a unique chimpanzee locus.

MANE select and protein coding transcripts are better connected between human and chimpanzee